Circle the correct answer for each question. If you're unsure, you can write numbers 1, 2, 3, etc. next to the choices to get partial credit.

A study published in 2007 by Christopher Johnson, professor of music education and music therapy at the University of Kansas, revealed that students in elementary schools who took private music lessons scored around 20 percent higher in math scores on standardized tests, compared to students who did not take private music lessons.

1. Is this a randomized experiment or an observational study?

   (a) Observational study, since math scores were not randomly assigned.
   (b) Observational study, since this was not a random sample.
   (c) Observational study, since whether the student took private music lessons was not randomly assigned.
   (d) Randomized experiment, since researchers gave students standardized tests.

Sickle-cell disease is a painful disorder of the red blood cells. To investigate whether drug A can reduce the pain associated with sickle-cell disease, a study by the National Institutes of Health randomly assigned 150 sickle-cell sufferers to receive the drug. Placebos were given to another 150 sickle-cell sufferers. Great care was used to ensure that the 300 participants did not know if their pill contained the drug. At the end of the treatment period, the researchers counted the number of episodes of pain reported by each subject.

2. Which of the following principles was NOT used in this study?

   (a) Control group
   (b) Blinding
   (c) Blocking
   (d) Randomization

Researchers randomly assigned patients with severe acne to one of two treatments: the standard topical antibiotic (the control) or a new antibiotic taken orally (the treatment). Six months later an evaluation of skin problems was made by a doctor who did not know which medication subjects had been given.

3. What is the primary advantage of random assignment of the treatment?

   (a) Random assignment allows the researchers to infer that any difference in outcome was due to the treatments.
   (b) Random assignment ensures that the study participants are representative of the larger population.
   (c) Random assignment ensures that the subject doesn't know which treatment they are getting.
   (d) Random assignment allows inference back to a population of patients with acne.

In a study on treatments for anorexia, researchers recruited 30 teenage girls with anorexia. The girls were ranked by weight prior to the study. Of the heaviest three girls, one was randomly assigned to a "family" therapy, one to a "cognitive/behavioral" therapy, and one to a control group. The same was done for the next three heaviest girls, down to the three lightest girls. After five weeks of treatment, researchers weighed the participants. Researchers would like to determine which, if any, of the treatments are most effective in treating anorexia. Use this scenario to answer questions 4 and 5.

4. If the researchers found that there were statistically significant differences in average weight across the three treatments, would the researchers be justified in concluding that the treatments caused the changes in weights?

   (a) Yes, since the therapy group was randomly assigned.
   (b) No, since weight was not randomly assigned.
   (c) Yes, since this was an observational study.
   (d) No, since the researchers do not know which treatment was most effective.

5. What type of study design did the researchers use?

   (a) Completely randomized design
   (b) Blocking study design
   (c) Matched pairs design
   (d) Observational study

Recently, a Montana newspaper asked people to go to the newspaper's website and respond yes or no to the question: "Should federal protections on grizzly bears be lifted?" Of 933 opinions, 58.1% said "Yes", and 41.9% said either "No" or "Uncertain".

6. Is it valid to conclude from the poll that over half of all Montana residents favor lifting the protections?

   (a) No, because the sample is too small given the population of 75,000 Montana residents.
   (b) Yes, because 58.1
   (c) Yes, because the sample size is large enough to represent all 75,000 Montana residents.
   (d) No, because the sample is not likely to be representative of the population.